Autoplay on Trial: An Online Experiment on the Suspect Behind Excessive Screen Time

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Abstract

Interface design can be strategically employed to 'nudge' consumers to take certain actions and is suspected of promoting addictive online behavior. A prevalent interface element across numerous popular social media and streaming platforms is the default autoplay feature. In this study, we present an incentivized online experiment to investigate whether the autoplay feature alone can cause an unwarranted increase in desired viewing times. While controlling for content, we demonstrate that the autoplay feature, in isolation, does not override participants' previously indicated preferences for media consumption. We discuss the our findings in terms of what actually renders online platforms addictive. Our results have direct policy implications for the proposed blanket ban on autoplay by the US Congress (SMART Act).

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1 Introduction

Autoplay is a user interface feature which automatically transitions users to new video content without any action. It has become a familiar design element in major social media and streaming platforms over the last decade,[1] and has been described as a default nudge that reduces the autonomy of users (Lukoff et al., 2021; Schaffner et al., 2025) Autoplay's widespread adoption and concerns about its restricting the freedom of choice has also attracted regulatory scrutiny, with the U.S. SMART Act proposing to ban platforms from serving autoplay content "without an express, separate prompt by the user" due to suspicions of its contribution to addictive online behavior.[2]

While autoplay is pervasive and tends increase viewing times (Chen et al., 2025; Hiniker et al., 2018; Schaffner et al., 2025), its impact on user welfare and the mechanisms driving this effect are not well understood. Does the increase in consumption go against users' wishes, or does it merely remove frictions to help them achieve their desired consumption levels? Can we attribute the increase in consumption to autoplay in isolation, or does it also depend on the supply of content? One reason for these gaps is the challenge of holding constant the video content displayed to users while eliciting their preferences related to content consumption before the actual consumption occurs.

We overcome this challenge in an incentivized two-day online experiment designed to isolate the causal effect of autoplay on content consumption. We recruited 184 participants who twice allocated 1200 seconds (20 minutes) between two tasks. Allocations were first made hypothetically using the strategy method, one day before the work date, and then realized on the work date. The tasks involved seconds of transcription of meaningless CAPTCHA texts at 0.38 pence/second, and seconds of watching funny animal videos at 0.33 pence/second. On the first day, participants familiarized themselves with

[1] As of 2025 these platforms include Facebook, Instagram, Netflix, YouTube, TikTok, and Twitter (currently X). Twitter first introduced autoplay in 2015, and described it as a means to reduce "extra effort" in the number of clicks/taps required to consume content: https://blog.x.com/official/en_us/a/2015/introducing-a-more-seamless-video-experience-with-autoplay.html.

[2] SMART Act of 2019, S. 2314, 116th Congress: https://www.congress.gov/bill/116th-congress/senate-bill/2314/text

both tasks in a practice session and made their preferred time allocation for the next day. We incentivized these allocations by randomly implementing 5% of their choices on the work date. On the next day, remaining participants allocated their time without constraints. Differences between planned and realized allocations allowed for the measurement of dynamic inconsistency, standard in intertemporal choice experiments (Ericson & Laibson, 2018). This two-day structure with hypothetical and realized choices formed the foundation for testing our preregistered hypotheses regarding autoplay.

To isolate the causal effect of autoplay on these allocation decisions, we randomly assigned participants to either the **Autoplay** condition where videos played continuously during the watching task or to the **Control** condition requiring manual clicks to play each video. Across conditions, we held constant the content by presenting 80 manually curated "funny animal" videos from YouTube and TikTok in identical sequence. We assigned participants to the same video interface setting on both days, i.e., participants experienced the same video interface setting on both practice and main sessions. This eliminated algorithmic content curation and personalization as potential confounds, ensuring that any observed differences in content consumption can be attributed to the autoplay setting. We dedicated the first block of the experiment to measure the differences between planned and realized allocations across conditions.

In a second block of the experiment, we modified the design to elicit demand for a commitment device. The commitment allowed participants to probabilistically favor their autoplay preference during 20 minutes on the main session. We recruited 52 additional participants who made nine binary choices between the autoplay setting (on or off) for the 20-minute main session and receiving bonus payments ranging from £0.05 to £0.50. One decision was randomly selected for implementation, determining both the autoplay setting for the main session and the associated bonus payment. This approach allowed us to measure participants' Willingness To Pay (WTP) for autoplay in advance, providing insight into whether participants viewed the feature as beneficial (positive WTP) or as a self-control problem requiring a commitment device (negative WTP).

We proceed with reporting our main findings. The first set of results focuses on our preregistered hypotheses regarding the increase in content consumption that we attribute to autoplay. In the main session, participants in the **Autoplay** condition watched statistically indistinguishable numbers of videos compared to the **Control** condition and the **Autoplay** condition had no significant effect on the proportion of time spent on either task. Similar results hold for other engagement metrics (i.e., average session length, number of sessions). To sum, we find no evidence that autoplay alone increased content consumption in our setting.

Second, contrary to our hypothesis that participants would demand commitment devices against autoplay, we find a positive WTP for the feature. Using data from the second block only, we are able to estimate an average WTP of 2.24 pence (about 45 seconds of typing) per 20-minute session. This implies participants viewed autoplay as a convenience feature rather than as a threat to their self-control which would have required a costly commitment device.

Third, comparing hypothetical time allocations to realized ones, we find that participants consumed significantly less content than their day 1 plans implied in both conditions. The decrease amounts to 15.8 percentage points (35%) less time spent on the watching task with day 1 plans as the baseline. This result is driven by 33% of all participants who had planned to spend up to three quarters of their time watching videos but instead decreased their content consumption on the next day. On the other end, 88% of those who had initially planned to spend less than a quarter of their time watching videos successfully followed their plans and did not succumb to temptation.

Supporting evidence from a regression discontinuity analysis reveals that participants monitored their day 1 plans despite ultimately exceeding them. When participants reached their planned typing duration, they immediately reduced typing probability by 25.5 percentage points in the next 60-second window. However, this effect dissipated quickly, becoming insignificant within 90 seconds, as participants returned to the typing task once again.

Taken together, these findings cast doubt on whether the watching task was actually perceived as tempting. A mixed-method analysis of open-ended responses reveals that participants perceived the experimental environment as a work setting rather than a choice between independent tasks. While 29% spontaneously described typing using negative terms, 23% characterized video-watching as "taking breaks" from work. This suggests that experimenter demand effects may have interfered with the intended effects of the autoplay manipulation, with implications for both the interpretation of our null results and the broader generalizability of our study.

A primary internal validity concern in online experiments is ensuring genuine participant engagement equivalent to laboratory conditions. We implemented several measures to address this challenge. We used JavaScript-based attention monitoring to detect when participants' browsers lost focus or when they navigated away from the experimental interface CITE INTERSECTION OBS PAPER. We required minimum internet speeds (30 Mbps on day 1, 25 Mbps on day 2) and restricted participation to desktop users with Chromium-based browsers to standardize the technical environment. Participants completed two attention checks and answered a post-experiment honesty question about their engagement while at the watching task. These controls ensured that participants in our final dataset genuinely allocated their full 20-minute attention to our tasks, approximating the controlled conditions of laboratory experiments while maintaining the scalability of online data collection CITE WORK.

Regardless, our task design may have inadvertently created experimenter demand effects that favored the typing task. While we structured payments to incentivize effort, this created a modest earnings difference of 60 pence over 20 minutes, equivalent to a 1.8 pound hourly wage difference (15% above our 12 pound minimum wage baseline). More problematically, we introduced quality criteria exclusively for the typing task and set typing as the default upon session initiation, both intended as internal validity measures. These design choices may have inadvertently signaled that typing was the "preferred" activity from the experimenters' perspective. Alternative designs could have addressed this concern by rewarding balanced time allocation, eliminating earnings altogether, or extending session duration beyond 20 minutes to increase the appeal of leisure breaks, though each approach presented trade-offs given our research objectives and budget constraints.

A major challenge in isolating autoplay effects is the role of content appeal. While we conducted pretests with UK adults from the same Prolific pool who selected animal videos as preferred content, and participants rated our curated videos positively (median 7/10, no difference across treatments), our manually selected content clearly lacked the personalized, algorithmically-optimized engagement of actual social media platforms. This represents the core trade-off in our experimental design: achieving causal identification of interface effects required holding content constant, but doing so necessarily reduced the ecological validity of the content experience that makes autoplay potentially problematic in practice. CITE WORK THAT ALLOWS SELECTION OVER SOME PRESELECTED CONTENT.

The theoretical mechanism through which autoplay might increase consumption draws on models of dual-system decision-making (Kleinberg et al., 2022). In this framework, consumption continues as long as the impulsive System 1 derives positive utility, with the reflective System 2 only able to intervene during natural breaks between content. Autoplay removes these decision points, preventing System 2 from regaining control and potentially trapping users in extended viewing sessions. Crucially, the success of the mechanism depends on content characteristics: if the content lacks sufficient appeal or "stickiness," System 1 receives negative utility and consumption stops regardless of autoplay. Our curated animal videos—while rated positively by participants (median 7/10)—may not have provided the personalized, algorithmically-optimized engagement found on actual platforms. This raises a fundamental question about policy interventions: if autoplay's effects depend critically on content characteristics, can interface-level regulations effectively address overconsumption without considering the broader ecosystem of recommendation algorithms, social features, and content curation?

Our experimental design addresses a key challenge in studying effects of user interfaces and separates the impact of design features from content characteristics. Previous studies either allow participants to choose their preferred content, conflating content preferences with interface effects, or conduct field experiments where users can easily circumvent restrictions by switching devices. We created a controlled environment where all participants viewed the same curated sequence of 80 short animal videos from YouTube and TikTok, eliminating algorithmic personalization and content variation. Participants faced real opportunity costs, earning 0.38 pence per second for typing tasks versus 0.33 pence per second for watching videos. This payment structure, combined with a cognitively demanding typing task (retyping distorted CAPTCHAs), ensured that time allocation decisions involved meaningful trade-offs between effort and entertainment.

Our study makes several contributions to understanding interface design effects on user behavior. First, we provide the cleanest test to date of autoplay's isolated impact by holding content constant across conditions—addressing a critical confound in prior work where treatment effects could stem from differences in either interface design or content exposure. Second, we document an important methodological insight: experimenter demand effects can reverse expected behavior in online experiments like ours. Participants viewed our experimental environment as a workplace setting where they should maximize productivity, leading them to type more than planned rather than giving in to tempting videos. This finding suggests that laboratory studies of digital "addiction" may systematically underestimate real-world effects when the experimental context signals that effort is valued. Third, our multiple price list treatment reveals that users have positive willingness to pay for autoplay, averaging 2.24 pence for a 20-minute session. This preference for autoplay as a convenience feature contradicts its characterization as a manipulative dark pattern that users would pay to avoid.

We contribute to several strands of literature examining digital consumption and self-control. Our null findings contrast with observational studies documenting excessive social media use and associated welfare losses (Allcott et al., 2020, 2022), suggesting that interface features alone may not drive these patterns. We extend the experimental literature on digital interventions in two important ways. Unlike field experiments that find large effects of usage restrictions (Beknazar-Yuzbashev et al., 2025), we can rule out substitution to other devices or platforms as an explanation for our results. Our design also improves upon laboratory studies that allow content choice (Ek & Samahita, 2023), which cannot distinguish whether effects stem from interfaces or from participants selecting more engaging content when barriers are lower. By holding content fixed, we show that when content lacks the "stickiness" of personalized feeds, autoplay has minimal impact—supporting theoretical models where interface effects depend critically on content characteristics (Kleinberg et al., 2022). Our findings align more closely with the literature on benevolent defaults (Carroll et al., 2009), where reducing decision costs improves user experience without distorting choices, than with concerns about manipulative design patterns that override user preferences.

Recent theoretical and empirical work has established that digital platforms can systematically exploit psychological vulnerabilities to increase user engagement. Allcott et al. (2022) develop an economic model of digital addiction and find that self-control problems account for 31% of social media use, with temporary incentives to reduce usage having persistent effects that suggest habit formation. This aligns with Ichihashi and Kim (2023), who show that platforms strategically choose service "addictiveness"—features that yield lower overall utility but higher marginal utility of continued attention—especially when competing for scarce user attention. As Loewenstein and Wojtowicz (2023) argue, attention has become a primary factor of production in the digital economy, making its capture economically valuable even when socially harmful. These models build on established evidence that individuals exhibit present-focused preferences in intertemporal choices (Ericson & Laibson, 2019), suggesting mechanisms like autoplay could trap users by exploiting the gap between immediate impulses and long-term preferences. Our experiment tests this specific mechanism in isolation, finding that autoplay alone fails to increase consumption when content lacks the algorithmic personalization and social features that make platforms genuinely "sticky." This null result suggests that while the theoretical models correctly identify vulnerability to digital addiction, the binding constraint may be engaging content rather than interface frictions.

Our two-day design, with participants stating preferences 24 hours before implementation, builds on a rich literature examining planning and self-control in laboratory settings. The experimental evidence on inconsistent planning presents a puzzle for standard models. Bhatia et al. (2021) find that when allocating time between work and leisure tasks in a laboratory setting, participants tend to plan improving sequences (delaying gratification) and prefer commitment devices, yet few actually deviate from their plans when given flexibility—and those who do tend to postpone leisure further rather than succumb to temptation. This contrasts sharply with field evidence of present bias and suggests that laboratory contexts may fundamentally alter behavior. Bonein and Denant-Boèmont (2015) demonstrate that self-control in experimental settings is influenced by social considerations: participants who commit to effortful tasks in the presence of peers maintain higher performance, suggesting that perceived social pressure can substitute for commitment devices. Our results extend these findings in an unexpected direction. Not only did participants fail to succumb to video temptation as models of present bias would predict, they systematically exceeded their planned typing time by 16 percentage points. This "reverse present bias" reveals how strongly experimental framing shapes behavior: participants interpreted our online laboratory as a workplace where effort signals virtue, overwhelming any temptation from our leisure task.

The question of whether interface features like autoplay represent benevolent choice architecture or manipulative "dark patterns" depends critically on alignment between designer and user interests. Traditional nudging, as conceptualized by Thaler and Sunstein (2009), aims to help people make better decisions for themselves while preserving freedom of choice. When applied by private firms, however, nudges may prioritize corporate rather than consumer welfare. Mills (2020) reintroduces the concept of "sludge"—frictions that make desirable behaviors harder—and argues that nudges and sludges exist symmetrically: reducing friction for one option necessarily increases relative friction for alternatives. Digital interfaces exemplify this tension. Caraban et al. (2019) identify 23 distinct nudging mechanisms in technology design, many of which can serve either user goals or platform profits. The evidence on autoplay specifically reveals this duality. While Lukoff et al. (2021) find that YouTube's autoplay undermines users' sense of agency and leads to "mindless" consumption they later regret, the effect appears strongest among vulnerable populations: Hiniker et al. (2018) show that autoplay ("post-play") significantly reduces preschoolers' ability to self-regulate screen time. Our results complicate this narrative. Adult participants not only resisted autoplay's supposed manipulation but expressed willingness to pay for it, averaging 2.24 pence per session. This suggests autoplay may function more like the optimal defaults studied by Carroll et al. (2009)—reducing decision costs without distorting choices—at least when content lacks the algorithmic personalization that makes platforms genuinely addictive.

Our findings underscore crucial methodological considerations for studying digital consumption behaviors. Zizzo (2010) identifies experimenter demand effects as a critical threat to validity in economic experiments, particularly when the experimental objectives align with socially desirable behaviors. This problem appears especially acute in digital consumption studies: participants may view reducing screen time as virtuous, leading them to behave differently in experimental settings than in their natural environments. The contrast between our null results and evidence from field studies illustrates this challenge. Beknazar-Yuzbashev et al. (2025) find substantial engagement increases when platforms reduce content moderation in a browser extension field experiment—but their design allows users to circumvent restrictions by switching to mobile devices, conflating treatment effects with platform substitution. Conversely, Ek and Samahita (2023) allow participants to choose their preferred YouTube content in a laboratory setting, finding evidence of self-control problems, but cannot distinguish whether effects stem from interface features or from participants selecting more tempting content when barriers are lower. The regulatory landscape further complicates empirical research. As Lupiáñez-Villanueva et al. (2022) document in their study for the European Commission, dark patterns rarely operate in isolation, often combining multiple manipulative elements within single interfaces. This clustering makes it difficult to isolate causal effects of specific features like autoplay. Moving forward, research designs must account for three key factors: the artificial virtue signaling inherent in laboratory studies of digital consumption, the need to separate interface effects from content effects, and the complex interplay between multiple design elements. Only through careful triangulation of laboratory experiments (which provide causal identification but may lack external validity) and field studies (which capture real behavior but face identification challenges) can we develop evidence-based policy for the digital economy.

The remainder of this paper is organized as follows. Section 2 describes our experimental design, including the two-day structure with practice and main sessions, the typing and video-watching tasks, and the multiple price list treatment for eliciting willingness to pay. Section 3 details our sample recruitment from Prolific and data collection procedures. Section 4 presents our results, beginning with participants' planned time allocations, their actual behavior, and the null effects of autoplay on video consumption. We also examine experimenter demand effects through qualitative analysis and test for behavioral changes around goal achievement. Section 5 analyzes participants' willingness to pay for autoplay from the multiple price list treatment. Section 6 discusses the implications of our findings for understanding digital addiction and policy interventions. Section 7 concludes.

2 Design

2.1 Overview

To explore the relationship between default video player settings and self-control, we designed a choice environment that captures the tension between productive effort and immediate gratification. Our experimental setup is centered around two competing tasks: A primary typing task that is tedious and offers limited learning opportunities, and a secondary video-watching task that provides immediate entertainment value. This design mirrors real-world situations where individuals must allocate time between effortful productive activities and readily available leisure alternatives.

The key feature in our design is the manipulation of autoplay setting in the video interface. Participants can switch between tasks at any time, but the ease of continuing with the leisure activity varies systematically across treatment conditions. This allows us to isolate the causal effect of interface design on time allocation decisions while holding constant individual time preferences and other task characteristics.

2.2 Timeline and Session Structure

We conducted a two-day online experiment to examine how autoplay features affect time allocation between productive and leisure tasks. The multi-day structure allowed us to measure the gap between stated preferences and actual behavior under two different video interface conditions. Participants completed two separate data collection sessions with a cooling-off period of 24 hours (up to 48 hours) between sessions.

Day 1 sessions included a practice session to familiarize participants with the experimental interface, followed by preference elicitation measures. Day 2 featured the main 20-minute work session where participants made actual time allocation decisions. We conducted two waves of data collection during May 2023, as described in Table 1.

Table 1: Summary of experimental design and timeline

	Practice Session	Time Choice	Main Session	Multiple Price List	Decision Horizon
Day 1 (May 10)	✓	✓			24-hour
Day 2 (May 11)			✓		immediate
Day 1 (May 18)	✓	✓		✓	24-hour
Day 2 (May 19)			✓		immediate

Note: This table shows the experimental components administered across two data collection waves, with columns ordered chronologically within each day. The first wave (May 10-11) established baseline treatment effects, while the second wave (May 18-19) incorporated the Multiple Price List treatment to measure demand for commitment devices. Decision horizon refers to the time between preference elicitation and the main experimental session.

2.3 Payment Structure and Incentives

We implemented a performance-based payment system to ensure participants faced meaningful trade-offs between tasks. We based payment directly on the seconds participants actively spent on each task, with different compensation rates for the typing and video-watching activities. This created clear opportunity costs for time allocation decisions while avoiding any completion bonuses that might distort behavior.

Participants who completed all required elements of the experiment received payment within seven days after the second session.[3] To maintain data quality, participants who failed to meet specified quality criteria for the typing task were not compensated for time spent on that task.

2.4 Technical Implementation and Interface Design

We designed the experimental platform to provide precise control over the testing environment while maintaining ecological validity. We presented both tasks within a single web page using distinct tabs, allowing participants to switch between activities seamlessly while preserving progress. The typing task always served as the default upon session initiation, requiring an active choice to switch to the video-watching alternative.

Our interface incorporated several key features to track behavior precisely. A progress bar and timer provided real-time feedback on session duration. We logged all critical interactions including tab switches, keystrokes during typing, video controls, and cursor movements outside the experimental window. This granular tracking enabled us to construct detailed measures of attention and engagement across tasks.

2.5 Internal Validity Controls

Online experiments present unique challenges for maintaining experimental control, including participant multitasking, technical disruptions, and environmental variation. We

[3] Prolific policy gives researchers 21 days to transfer funds to participants. The typical delay between study completion and payment is however only a couple of days on average, as indicated on the Prolific website (link).

implemented comprehensive measures to address these concerns and ensure data quality.

Participant and Technology Screening: We selectively admitted participants using multiple screening criteria to ensure stable experimental conditions. Internet speed requirements filtered out participants with unstable connections (minimum 30 Mbps on day 1, 25 Mbps on day 2). We restricted participation to users with Windows, Linux, or Mac operating systems using Chromium-based browsers exclusively. We excluded Safari and Mozilla Firefox due to restrictive autoplay configurations. Mobile devices and tablets were prohibited to standardize the user experience.

Platform Controls: The experimental website enforced linear navigation, preventing participants from moving backward through the experiment. We disabled right-click functions and keyboard shortcuts. Any attempt to navigate backward reset progress to the landing page. We clearly communicated these restrictions to participants at the experiment's outset.

Attention and Engagement Monitoring: We implemented real-time tracking of participant attention using JavaScript packages to detect interface visibility disruptions.[4] This system captured actions such as minimizing the browser, switching between web pages, or launching other applications. The tracking provided second-by-second accountability of participant engagement.

Post-Experiment Validation: After completing the experiment, participants reported any connectivity disruptions and confirmed their genuine engagement with the video-watching task. We combined these self-reports with our objective tracking measures to identify and filter data potentially compromised by distractions or divided attention.

2.6 Tasks

The experiment centers on participants' time allocation decisions between two distinct activities that differ substantially in their cognitive demands and entertainment value. Both the Typing and Watching Tasks were presented to participants twice: First during

[4] For more information about the package, see here.

the Practice Session to familiarize participants with the interface and task mechanics, and second during the Main Session where performance determined compensation. In both sessions, depending on the randomized assignment to conditions, the key experimental manipulation occurs in the Watching Task.

2.6.1 Typing Task

The Typing Task requires retyping successive CAPTCHAs. CAPTCHAs are computer-generated images that contain random uppercase and lowercase letters as well as randomly placed white spaces.[5] Each CAPTCHA has a total length of 35 characters, including the white spaces.

We designed the images to be blurred and distorted by lines and dots added on top to increase comprehension difficulty.[6] The interface always presents these CAPTCHA images with a text box and a submit button. When the submit button is clicked, the next CAPTCHA image appears and the text input for the previous one is stored. The task is identical across all conditions, as illustrated in Figure 1.

We established easy-to-meet quality requirements to ensure participants remained active. These requirements include an overall typing accuracy of at least 70 percent in submitted CAPTCHAs[7] and a minimum of one CAPTCHA submission per minute spent on the task.

Participants must meet these quality requirements to receive compensation for the Typing Task. We remunerated the Typing Task at 0.38 pennies per second participants worked on it. A hidden timer recorded each second participants kept the Typing Task in the visible part of their screen. Participants in both conditions spent on average 41

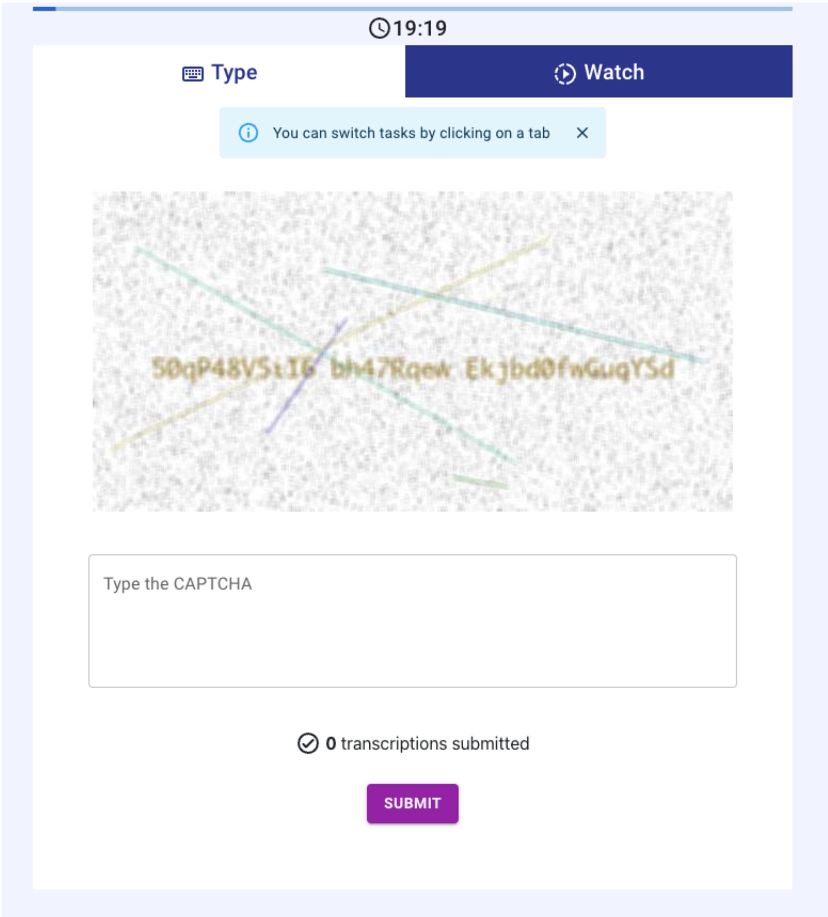
[5] White spaces were added to improve readability after pretests. The number of white spaces was limited to a maximum of 5 instances per CAPTCHA to control the typing time per image.

[6] The randomization procedure to generate CAPTCHAs is available here.

[7] We use Python's `difflib` string-matching algorithm to measure accuracy. It consists of finding the longest common substring and then finding recursively the number of matching characters in the non-matching regions on both sides of the longest common substring. Overall accuracy of 70 percent implies an average of 10.5 $(0.3 * 35)$ mistakes across all submitted CAPTCHAs.

seconds per CAPTCHA.

Figure 1: Typing Task tab



Note: This figure describes the typing interface tab. As participants always begin in the "Type" tab by default, there is a banner that reminds participants to click the tab buttons to switch tasks. This banner disappeared as soon as participants switched tasks once.

2.6.2 Watching Task

In this task, participants watch successive short videos in a customized media player. Since we are interested in isolating the media player's default settings, the video content and the order in which videos appear to participants are identical across conditions. We completely disabled the media player controls: Participants cannot skip videos (user scrolling is disabled) or use the media player's slider to advance or go back in a particular video. We remunerated the Watching Task at 0.33 pennies per second participants worked on it.

The conditions differ in their autoplay functionality. In the **Autoplay** condition, videos play automatically when the Watching tab is visible and pause with a click anywhere on the video. While the Watching task is visible, the media player keeps scrolling to the next videos and continues playing them unless interrupted. In the **Control** condition, participants need to have the Watching tab visible and click on each consecutive video to play them. At the end of each video, the scrolling animation brings the next video, but the new video only starts playing when the participant clicks on it. Figure 2 illustrates the video tab in the **Control** condition.

We curated videos from YouTube and TikTok, using tags such as "funny animal videos/short" and "cute animal videos/shorts". We manually selected 80 videos to isolate the difference in the media player design and help avoid algorithmic biases resulting from the recommender system's personalized suggestions. We chose animal videos because of their popularity during pilot studies.[8] We reviewed the selected videos to ensure that they did not contain any harmful or violent actions toward the animals involved. We kept the first 60 videos shorter in the beginning to maximize the frequency of experiencing the autoplay feature during the watching session, with videos getting longer for the last 20 videos (see Table 2).

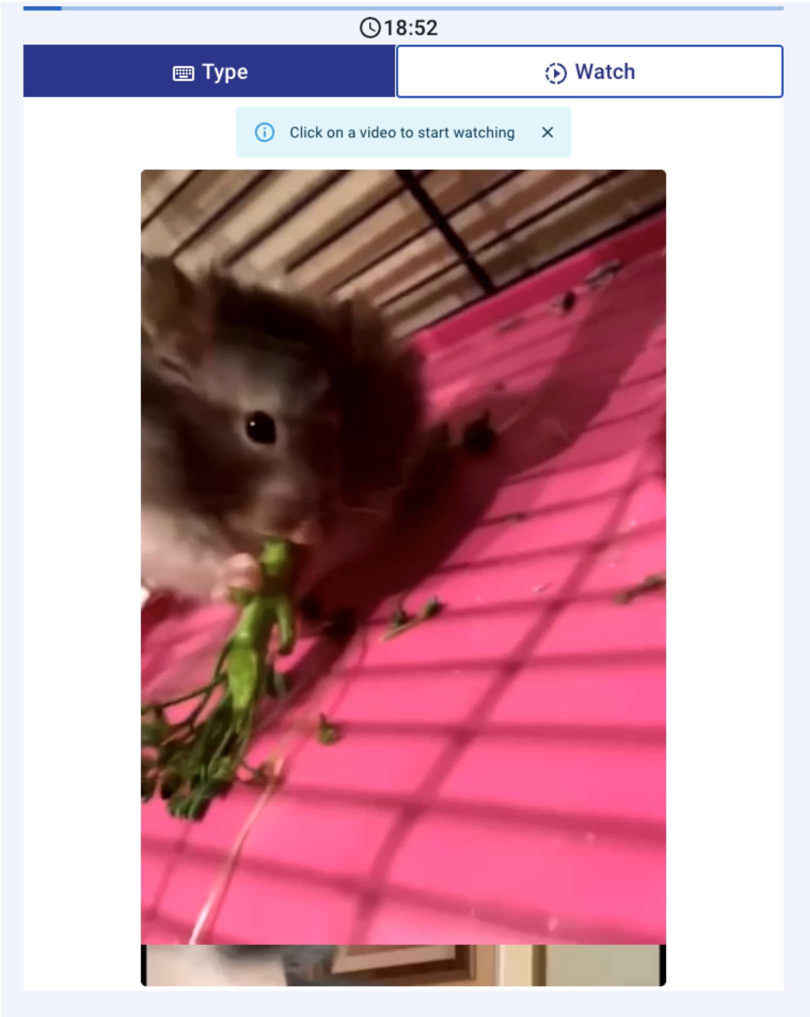
[8] During the pilots, the experiment was followed by a separate questionnaire asking participants to rate the videos they had seen and suggest what types of videos they would have liked to see more of.

Table 2: Video durations in the watching task

	First 60 videos	**Last 20 videos**
Duration (seconds)		
Mean (SD)	12.25 (7.04)	25.50 (14.04)
Median [Min, Max]	10 [5, 38]	19.5 [8, 60]

Note: This table presents the duration characteristics of the 80 animal videos used in the experiment. The first 60 videos were kept shorter to maximize participants' exposure to the autoplay feature during watching sessions, while the last 20 videos were longer.

Figure 2: Watching Task tab



Note: This figure describes the “Watch” tab in the **Control** condition. Before the first video is clicked upon to be played, there is a banner that reminds participants that the video won’t play without clicking. This banner was not present in the **Autoplay** condition and videos started playing as soon as participants clicked on the “Watch” button seen up on the right. Notice how top part of the following video is already visible in the bottom part of the screen.

2.7 Practice Session

The experiment consists of two sessions during which participants interact with the tasks described above. The first session is the Practice Session on day one. It comes right after the written descriptions of each task and serves to familiarize participants with the tasks and the interface. Participants are assigned to either **Autoplay** or **Control** condition before the Practice Session, so that they experience the same video interace on both the Practice and Main Sessions.

During the practice session, each task was available for only 60 seconds and participants were not allowed to switch between tasks. On the first tab, participants were required to retype CAPTCHAs. The 60-second period was enough to submit one CAPTCHA and see the second CAPTCHA image. After 60 seconds, a pop-up appeared on the screen blocking any other interaction and taking participants to the second page.

Participants were required to close this pop-up to advance to the second task. The rationale behind the pop-up was to ensure participants spent equal time on both tasks. Whether or not this pop-up was closed was verified to check participant activity and served as an additional attention check.[9] Once closed, the Watching Task began for 60 seconds, enabling participants to watch 4 to 5 short videos depending on whether they were assigned to the **Autoplay** or the **Control** condition. In the **Autoplay** condition, videos played continuously, while in the **Control** condition, videos required manual start.

We did not remunerate participants for the time they spent here because the purpose of the practice session was to understand the difficulty and pacing of the tasks, as well as navigation within the interface.

2.8 Multiple Price List

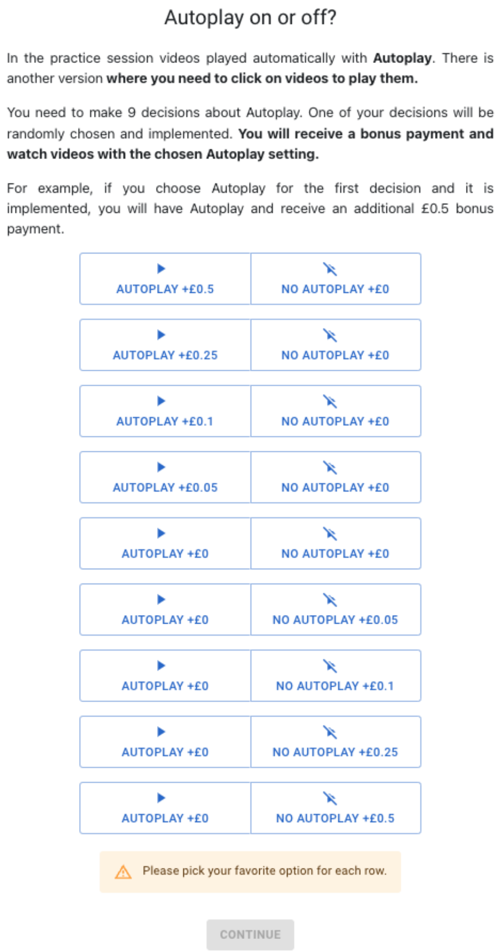
After the Practice Session on day one, participants in the Multiple Price List (**MPL**) condition made 9 binary decisions (see Figure 3). They chose between their preferred media player setting (Autoplay off or on) and a bonus payment. We introduced the MPL condition after the initial data collection phase. We determined its bonuses by the per-second earning differences across tasks, multiplied by the observed time differences between the Autoplay and control groups.

[9]All participants closed this pop-up and switched to the Watching Task.

Specifically, the average difference between the Autoplay and control treatments, in terms of time spent across tasks, was approximately 60 seconds (as shown in Table 8). This resulted in an average earning difference of $60 * (0.38 - 0.33) = 3$ pennies. To capture this in the MPL choices offered, we set the symmetric bonuses accordingly: £$[0.05, 0.1, 0.25, 0.5]$.

Once these decisions were completed, participants received information on the media player setting for day two—specifically whether Autoplay would be on or off—and the associated bonus. This information was displayed on the next page. With this knowledge, they then decided on their Time Choice. Participants who were not in the MPL condition proceeded directly to the Time Choice page after their Practice Session. More details on the Time Choice can be found in the subsequent section.

Figure 3: Multiple Price List page



Note: This figure describes the MPL page where participants made 9 binary choices between their preferred autoplay setting and bonus payments. Each row presents a choice between "Autoplay" (left) and "No Autoplay" (right) with associated bonus amounts ranging from £0.05 to £0.5. The interface required participants to select one option in each row before proceeding. One of these 9 decisions was randomly selected for implementation in the Main Session, determining both the autoplay setting and any bonus payment for day two.

2.9 Time Choice

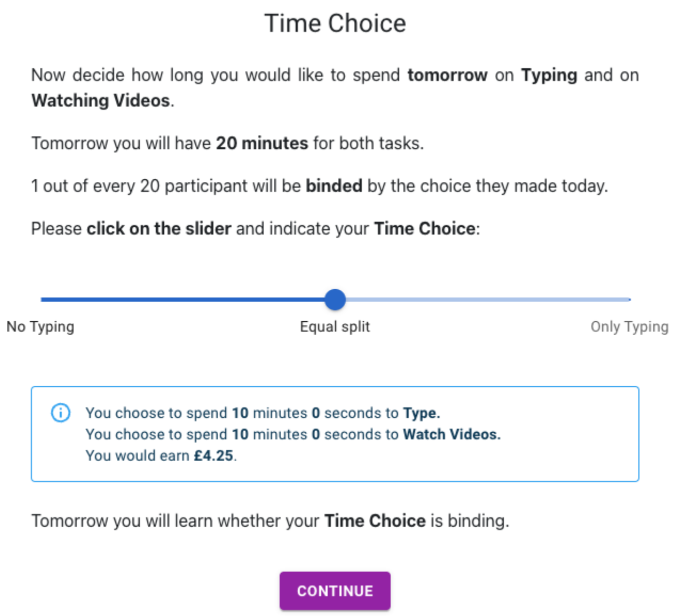
Following the Practice Session, participants transitioned to the Time Choice section. This section required them to specify their preferred time allocation between the two tasks for the main 20-minute session scheduled for the following day. Participants knew the main session would be 20 minutes long and they were free to choose corner solutions, even if that meant focusing solely on one task.

We designed a slider tool with 20-second steps for participants to indicate their time allocation preferences. Initially, the slider appeared grayed out and required a click to activate. This design aimed to ensure participants considered their choices in an active manner. The click-to-activate design had two benefits: It guaranteed participants interacted with the slider and helped participants understand the task-related payoff structure.

To emphasize the importance of this decision, we made the choices of 5 percent of participants binding in the main session. Binding choices led to payments based on the initial Time Choice, but only if participants met the quality standards described under Section 2.6.1. We excluded data from these participants from the final dataset accordingly. We informed participants whether their allocation was binding with their Time Choice reminder on day two, right before the start of the main session.

The Time Choice served as a benchmark for participants, and participants accordingly received a textual reminder of it on day two. The Time Choice section offers insights into participants' informed time preferences allows gauging their commitment to these choices after the 24-hour cooling period.

Figure 4: Time Choice page



Note: This figure shows the Time Choice interface with a slider tool allowing participants to allocate time between the two experimental tasks for the following day's 20-minute Main Session. The slider initially appears grayed out and requires a click to activate. Real-time feedback shows the selected allocation (here, 10 minutes each) and expected earnings (£4.25). The interface informs participants that 1 out of 20 (5 percent) will be bound by their choice and that they will learn whether their choice is binding on day two.

2.10 Main Session

The Main Session took place on day two, approximately 24 hours after the Time Choice. We began by reminding participants of their previous day's allocation and informing them whether their choice was binding or whether they were free to allocate their time as they preferred.

The session lasted exactly 20 minutes. Participants could freely switch between the Typing and Watching Tasks at any time using the tab interface. Unlike the Practice Session, there were no time restrictions on individual tasks, and participants had complete autonomy over their time allocation decisions. The interface maintained real-time tracking of time spent on each task, and participants could monitor their progress and potential earnings through the progress bar and timer.

We implemented the experimental treatment during this session through the media player's autoplay functionality. Participants assigned to the **Autoplay** condition experienced automatic video playback when the Watching tab was visible, while those in the **Control** condition required manual clicks to start each video. All other task mechanics remained identical to the Practice Session.

Payment was calculated based on actual time spent on each task, with compensation rates of 0.15 pence per second for Typing and 0.1 pence per second for Watching. Participants needed to meet the established quality requirements for the Typing Task to receive compensation for time spent on that activity. The session concluded automatically after 20 minutes, and participants received their final payment calculation before completing the experiment. Participants received a fixed completion fee of £2.75. Average earning per participant was £4.08, amounting to an average payment of £1.33 (median £1.49) for the main session.

2.11 Hypotheses

Kleinberg et al. (2022) model user consumption as driven by two competing systems: System 1 (impulsive, myopic) and System 2 (thoughtful, forward-looking). System 2 represents users' "actual" preferences, while System 1 drives continued consumption even when users derive little utility. Content exists within a *content manifold* that can be characterized along three dimensions: how much users genuinely value the content, how long they naturally want to engage with it, and how "sticky" or addictive it is akin to the difference between nutritious salad and moreish potato chips.

Following this framework, autoplay reduces friction in content consumption by removing decision points between videos, making content more "sticky" and decrease the likelihood that users' thoughtful system will regain control between consumption decisions. The key assumption here is that the experimental videos have sufficient baseline appeal and stickiness to allow autoplay to meaningfully increase engagement. As Kleinberg et al. (2022) demonstrate, UI design choices are "not separable from other content choices, since their effect depends on where the platform has positioned its content on the underlying content manifold." If content is purely "salad-like" with very low inherent appeal and stickiness, autoplay may have minimal effect on video consumption. This is analogous to how introducing breaks would have little impact on engagement for low-stickiness content like documentaries, but dramatically reduce engagement for high-stickiness content like celebrity gossip. With the assumption that our selection of videos have sufficient appeal and stickiness, **Autoplay** condition should lead to consumption beyond users' stated preferences, creating a measurable gap between intended (Time Choice) and actual behavior. In this case, we expect autoplay's behavioral effects to be known by participants, albeit underestimated as to how much they would be affected by these interfaces (Bongard-Blanchy et al., 2021). We thus expect positive demand for a comittment device driven by "sophisticated" participants in line with time preference literature.

Hypothesis 1: **Autoplay** condition decreases time allocated to the typing task while increasing video consumption. We test this using a linear regression:

$$Y_i = \alpha + \beta \text{Autoplay}_i + \gamma \text{TimeChoice}_i + \delta(\text{Autoplay}_i \times \text{TimeChoice}_i) + \epsilon_i \qquad (1)$$

where Y_i represents our outcome variables of interest (proportion of time spent typing or number of videos watched), Autoplay_i is a binary indicator for the treatment, and TimeChoice_i is the standardized proportion of time participants planned to spend typing on day 1. We estimate this model with and without the interaction term to examine whether the autoplay effect varies based on initial planning preferences. We expect $\beta < 0$ for typing and $\beta > 0$ for videos watched.

Hypothesis 2: **Autoplay** condition has an effect on deviations from stated time allocation preferences (actual minus planned time spent on typing). We test this using a two-sample t-test comparing the means between the two groups.

Hypothesis 3: If participants are aware of their self-control issues, they should exhibit positive Willingness To Pay (WTP) for removing autoplay, preferring the commitment devices that restricts automatic video transitions. We test this using a logistic regression:

$$\Pr\left(\frac{\text{Autoplay}_{ij}}{1-\text{Autoplay}_{ij}}|\text{Bonus}_{ij}\right) = \Lambda(\alpha + \beta\text{Bonus}_{ij}) \tag{2}$$

where Autoplay is the binary choice of taking autoplay given the bonus j from the MPL for individual i. We calculate the WTP for autoplay as $-\alpha/\beta$, and expect a positive sign indicating preference for commitment against autoplay.

In sum, these hypotheses test whether interface design features can systematically distort optimal time allocation decisions in a setting where participants face clear monetary incentives favoring productive work.[10]

3 Sample and Procedures

The experiment was run entirely online, using widely available open-source tools.[11] We drew the sample from the Prolific participant pool, an online recruitment platform commonly used for academic research, targeting adult British residents. We further imposed the following criteria: Fluency in English, balanced gender representation, and having a stable internet connection (> 30 Mbps on the first and > 25 Mbps on the second day of the experiment). We did not allow mobile devices and restricted access to the experiment to Chromium-based browsers to enhance the internal validity of the experiment as previously discussed in Section 2.5. Only participants who succeeded on both attention checks, reported not having engaged in other activities, and did not have connection issues during the experiment were included in the final dataset.

For the first stage of data collection, that is to say excluding the MPL treatment, we preregistered a two-sample parallel design using our prototype data and tested for equality of means across treatments. We found that the required sample size to test

[10] To access the preregistered hypotheses, click here. Our preregistration included hypotheses 1 and 3. We have added supporting hypothesis 2 regarding our expectations about the Time Choice section later.

[11] We used the MERN stack to develop and host the website in an entirely free way.

Hypothesis 1 with the usual parameter values ($\alpha = .05$, $\beta = .2$) would be 114 per treatment, with an effect size of 65 seconds calculated by the difference in the average time spent between conditions ($\mu_{Autoplay} - \mu_{Control}$) and population standard deviation of 165 seconds. Since attrition in longitudinal experiments is commonplace, we invited 301 participants on the first day. On day two, we invited the same 301 participants and received 276 complete responses. No participants failed the attention checks more than twice, and therefore were not removed from the dataset as per Prolific policy. We dropped 17 participants who self-reported having connection issues and 24 additional participants who stated having engaged in other activities. We also removed 11 participants who had their Time Choice enforced randomly. Finally, we removed 40 participants that were detected to spend more than 20 consequtive seconds away from our experimental platform who disproportionately come from the autoplay condition. These 40 participants all reported engaging with videos but had their mouse detected outside the experiment window, making it impossible to distinguish whether they genuinely watched videos or browsed other tabs/windows during the study (see Appendix Table A.1). The final dataset on which we base the analysis consisted of 184 individuals. Table 3 shows the demographic characteristics of the observed individuals.

Table 3: Balance table for the sample

	Autoplay N = 79	**Control** N = 105	***p*** [A = C]
Gender			
Female	48 (46.15%)	54 (45.00%)	0.510
Age			
Mean (SD)	40.56 (11.66)	40.24 (10.98)	0.725
Employment			
Employed (full time)	62 (59.62%)	67 (55.83%)	0.718
Employed (part time/self)	28 (26.92%)	28 (23.33%)	0.417
Not employed	11 (10.58%)	18 (15.00%)	0.167
Income			
Low (0-15)	28 (26.92%)	34 (28.33%)	0.989
Middle (15-30)	30 (28.85%)	36 (30.00%)	0.505
High (50+)	15 (14.42%)	13 (10.83%)	0.325
Marital Status			
Married/Partnership	56 (53.85%)	60 (50.00%)	0.510
Single	42 (40.38%)	52 (43.33%)	0.534
Previously married	6 (5.77%)	8 (6.67%)	0.927

Note: This table presents demographic characteristics for the sample of 184 participants across treatment conditions. Income categories are presented in thousands of British pounds (£). Employment categories: "Not employed" includes unemployed (looking and not looking) and homemaker. "Previously married" includes divorced, widowed, and separated. p-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample t-tests. For employment, part-time and self-employed are together as one category. The sample demonstrates successful randomization with balanced participant characteristics across the conditions.

For the MPL treatment, applying the same selection criteria as before, we invited

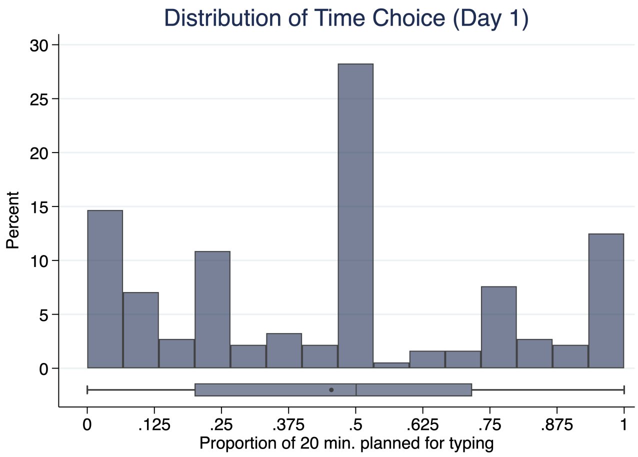
60 participants on day one. On day two, we received 57 complete answers. Once again, no participant failed the attention checks. No participant reported having connection issues. This improvement is likely because we had implemented higher network speed requirements (> 35 Mbps on the first and > 30 Mbps on the second day). We dropped 5 participants who reported having engaged in other activities. As we were interested in the demand for a commitment device, participants did not have their Time Choice enforced in this treatment. We therefore ended up with 52 individuals in the final MPL dataset.

4 Results

4.1 Time Choice: Planned time allocation on day 1

How did participants plan to allocate their time on day 1? We look at the participants' Time Choices to answer this question. These were their incentivized time allocation choices based on how much they would like to spend on the Typing task for the 20-minute main session 24 hours before. Figure 5 illustrates how the Time Choices of participants were distributed along 5 focal points: The modal action for about a third of all participants, was choosing to divide their time equally between the two tasks, then another third chose to only type or watch videos, with 15% choosing typing only and 15% choosing watching only. The remainder of participants were distributed around either 1/4 of the time typing, or 3/4 of the time typing. Overall, the day 1 decisions can be characterized as symmetrical around dividing time across either task.

Figure 5: Distribution of Time Choice



Note: This figure shows the distribution of participants' Time Choices for the 20-minute Main Session, measured as the proportion of time planned for typing. The histogram reveals clustering around five focal points: Approximately 20% chose all watching, 12% chose 25% typing, 29% chose equal time allocation, 9% chose 75% typing, and 14% chose all typing. The box plot below shows the overall distribution with median at 0.5.

Based on the observed distribution of the Time Choices, we categorize the day 1 intentions of participants into three distinct groups: Those who almost always planned to type, those who almost always planned to watch videos, and those who planned to mix between the two tasks in various proportions. Table 4 summarizes each group.

Table 4: Participant types by planned time allocation on day 1

	Mostly watching	**Alternate**	**Mostly typing**
Time Choice			
Share	28.3%	53.8%	17.9%
Mean (SD)	0.07 (0.08)	0.49 (0.15)	0.95 (0.07)
Median [Min, Max]	0.00 [0.00, 0.23]	0.50 [0.25, 0.75]	1.00 [0.78, 1.00]

Note: This table presents the clustering of participants based on their planned time allocation choices for the 20-minute Main Session on day 1. Watching-focused participants planned to spend 25% or less of their time typing, alternating participants planned to spend between 25% and 75% of their time typing, and typing-focused participants planned to spend more than 75% of their time typing.

4.2 Main Session: Actualized typing decisions on day 2

How did participants allocate their time during the 20-minute main session on day 2? Because we collected second-by-second data on participant behavior, we can track how they allocated their time between the two tasks. Based on the observed distribution of the actualized time spent typing, we categorize the day 2 behavior into three groups following the same rule: Those who almost always typed (more than 75% typing), those who almost always watched videos (less than 25% typing), and those who mixed between the two tasks in various proportions. Table 5 summarizes each type. We observe that the largest share of participants are now either alternating or mostly typing.

Table 5: Participant types by actualized time allocation on day 2

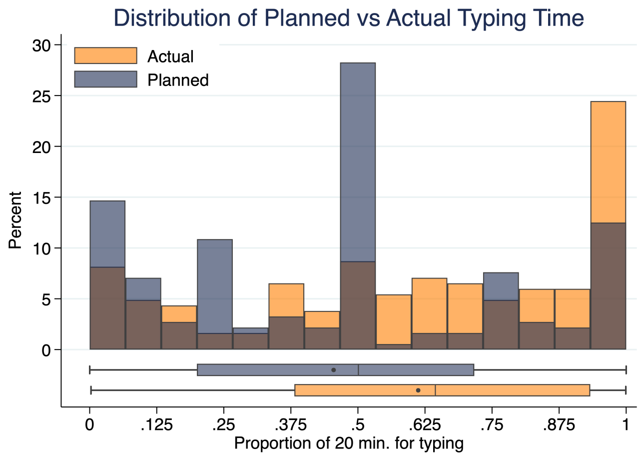
	Mostly watching	Alternate	Mostly typing
Type Time			
Share	19.3%	39.8%	40.9%
Mean (SD)	0.09 (0.08)	0.54 (0.13)	0.94 (0.08)
Median [Min, Max]	0.08 [0.00, 0.24]	0.52 [0.28, 0.74]	0.97 [0.75, 1.00]

Note: This table presents the clustering of participants based on their actualized time allocation during the 20-minute Main Session on day 2. Type Time represents the proportion of time actually spent typing. Mostly watching participants spent 25% or less of their time typing, alternating participants spent between 25% and 75% of their time typing, and mostly typing participants spent more than 75% of their time typing.

4.3 Discrepancy between planned (day 1) and actualized (day 2) time spent on typing

How do participants' day 1 time allocation compare to their actualized time spent on day 2? Participants spent more time on typing than they anticipated, even though we reminded participants of their day 1 plan, and the entire earnings difference between typing for 20 minutes versus watching videos was set to be 60 pence (£0.60). On average, participants planned to allocate 45.5% of their time to typing but actually spent 61.2% of their time on this task, representing an increase of 15.8 percentage points ($t = -8.24$, $p < 0.001$). This extends beyond central tendencies and the entire distribution shifted rightward, with median actual typing time (64.3%) substantially exceeding median planned time (50.0%). The Kolmogorov-Smirnov test confirms that planned and actual distributions differ ($D = 0.86$, $p < 0.001$), indicating systematic deviations from day 1 intentions.

Figure 6: Distribution of Planned vs Actual Time Allocation



Note: This figure compares the distribution of participants' planned time allocation (Day 1) with their actual time spent typing (Day 2), measured as the proportion of the 20-minute session. The rightward shift in the actual distribution demonstrates participants' tendency to spend more time on the effortful typing task than originally planned.

To account for the substantial heterogeneity in both day 1 and day 2 decisions, we compare transitions from participants' day 1 type to their day 2 type in Table 6. We categorize participants into three behavioral types based on their time allocation: mostly watching ($<$ 25% typing), alternate (25-75% typing), and mostly typing ($>$ 75% typing). Overall, 63.0% of participants remained in the same behavioral category between planning and implementation, indicating moderate consistency in behavioral types despite the shift in continuous measures.

Mostly watching and alternate types showed the least adherence to their day 1 plans (59.6% vs 56.6% respectively), with both groups showing upward shifts toward more typing. Among mostly watching planners, 40.4% moved to higher typing categories (30.8% to alternate, 9.6% to mostly typing). Similarly, 40.4% of alternate planners shifted to mostly typing, while only 3.0% moved down to mostly watching. In contrast, mostly typing planners exhibited the highest stability at 87.9%. A chi-square test confirms that deviation rates differ significantly across initial behavioral types ($\chi^2 = 109.00$,

$p < 0.001$), indicating that initial behavioral intentions are strong but heterogeneous predictors of subsequent implementation patterns.

Table 6: Transition matrix: Planned vs actual behavior types

	Planned (Day 1)		
Actual (Day 2)	Mostly watching	Alternate	Mostly typing
Mostly watching	59.6%	3.0%	3.0%
Alternate	30.8%	56.6%	9.1%
Mostly typing	9.6%	40.4%	87.9%

Note: This table shows transitions between planned behavior types (Day 1) and actual behavior types (Day 2). Mostly watching (< 25% typing time), Alternate (25-75% typing time), Mostly typing (> 75% typing time). Diagonal entries show participants who remained consistent (63.0% overall). Off-diagonal entries show deviations: mostly watching planners (40.4% deviated), alternate planners (43.4% deviated), and mostly typing planners (12.1% deviated).

This represents a departure from traditional models of time inconsistency and present bias, which typically predict that individuals will succumb to immediate temptation rather than adhere to effortful plans. Instead, participants demonstrated what might be characterized as 'reverse present bias' and systematically allocated more time to the cognitively demanding task than originally intended. The robustness of this finding across the distribution suggests that factors beyond self-control failures govern planning and its implementation in our setting, and potentially include how both tasks and the decision environment were perceived in the experiment.

4.4 Experimenter demand effects

Why did participants end up typing more than their day 1 plans? To bridge the gap between our theoretical view of the experimental setup and how participants perceived it, we asked how participants decided to allocate their time during the 20-minute main session once the session was concluded. We explore these by analyzing participants' qualitative answers to an open-ended question on how they decided to divide their time between the two tasks using dictionary-based concept detection methods reviewed by Ash and Hansen (2023).

We employ dictionary-based concept detection methods within a bag-of-words framework to analyze participants' qualitative responses. Following standard approaches in computational text analysis, we preprocess the text data by converting to lowercase and apply pattern matching to identify specific term sets that relate to negative perceptions of the typing task and breaks towards the watching task. Using regex pattern matching, we create binary indicators for documents containing predetermined words associated with task difficulty (e.g., "tiresome","frustrating","difficult") and break-taking behavior (e.g., "break","rest","relax") - with the exception of "switch" which typically described simple task changing behavior without the implication of taking a break from typing. This approach allows quantifying the prevalence of common conceptual themes across participants' obligatory open-ended responses. We match algorithmic output with a manual review to confirm our results about participant perceptions. Figures 7a and 7b summarize these words by their frequency.

Figure 7: Word clouds for participant perceptions

(a) Typing task negative descriptions

(b) Changing to watching task

Note: Panel (a) shows words describing negative perceptions of the typing task. Panel (b) shows words describing switching to the watching task as break-taking behavior. Word size corresponds to frequency of mention across responses.

Our mixed methods analysis reveals that when asked to describe how they decided to switch between the tasks, 29% of participants described the typing task negatively at their own will, i.e., without being asked to comment on the typing task itself, while 23% specifically described watching videos as a break from the typing task. These findings suggest that the decision environment was perceived by many as a work environment, where one should type as much as possible while taking as few breaks as possible. The observed increases across all but always typing types, along with the qualitative perceptions suggest that experimenter demand effects have effected participant behavior.

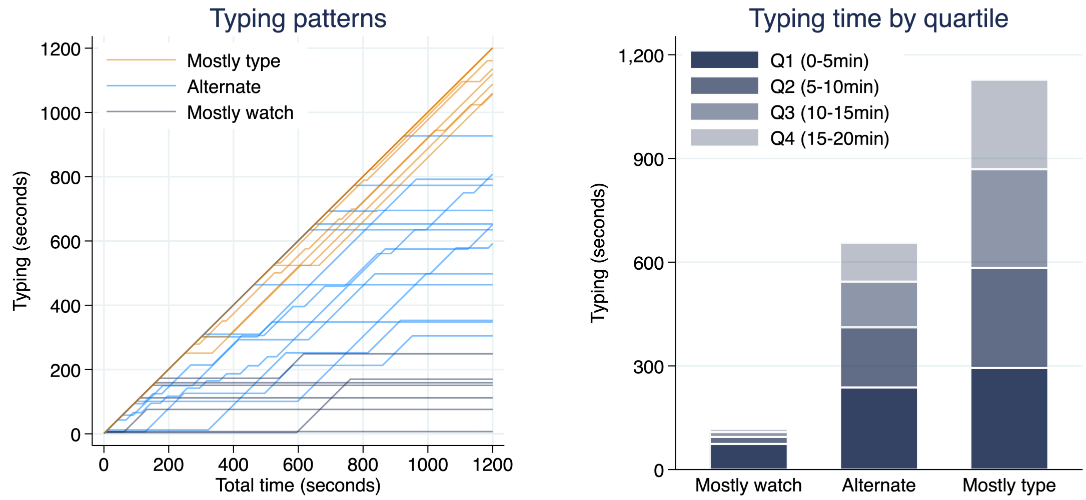
4.5 Effect of achieving day 1 typing objective

One test for this hypothesis is looking at what participants do around the window of achieving their day 1 goals. If their behavior changes, i.e., they start typing less after achieving their goal, this would indicate adherence to day 1 goals and limited distortion due to experimenter demand effects. Should they continue typing as if nothing happened, we should be worried of severe problems in the desing of the experiment. We test this exploratory hypothesis in the next section.

We categorize participants in their actualized typing time on day 2 using the same three categories from day 1 (see Figure 8a). For participants who are "alternating", typing periods are interrupted by watching periods, following with the "taking breaks" analogy made by participants. We observe substantial individual heterogeneity in lengths of typing and watching periods, and divide the main session across 5-minute periods to understand typing to watching proportions of participants. We find that typing periods get shorter and shorter, suggesting changing between tasks may be strategic as participants get tired of typing or achieve their day 1 goals (see Figure 8b).

Figure 8: Time spent typing

(a) Individual typing patterns (b) Average typing by quartile



Note: Panel (a) shows typing time for 30 random participants across the 20-minute session. Panel (b) shows average typing time by 5-minute periods, revealing declining typing duration over time.

Why do participants reduce time spent typing as they advance? If they are typing without regard to their day 1 objective, their typing shouldn't be affected by attaining their day 1 goal, and reduce gradually as fatigue or frustration grows from doing the typing task. Instead, if they care about hitting their goal, we should see a sharp drop in time spent typing right after attaining it.

To examine behavioral changes as participants achieve their day 1 plans, we implement a sharp regression discontinuity design using the moment participants reach their planned typing duration as the cutoff. We aggregate binary task choices (whether typing or watching) into 30-, 60-, and 90-second bins around this threshold and estimate the discontinuous change in the probability of choosing the typing task. Results in Table 7 reveal a significant negative effect immediately after plan achievement: participants are 27.4 percentage points less likely to continue typing in the 30-second window following plan completion (robust estimate: -0.274, p = 0.007) and 25.5 percentage points less likely in the 60-second window (robust estimate: -0.255, $p = 0.003$). However, this effect dissipates over longer horizons, becoming statistically insignificant in the 90-second window (robust estimate: -0.075, $p = 0.487$). These findings suggest that achieving initial plans triggers an immediate behavioral shift toward leisure consumption, consistent with participants treating plan completion as a license to reduce effort temporarily.

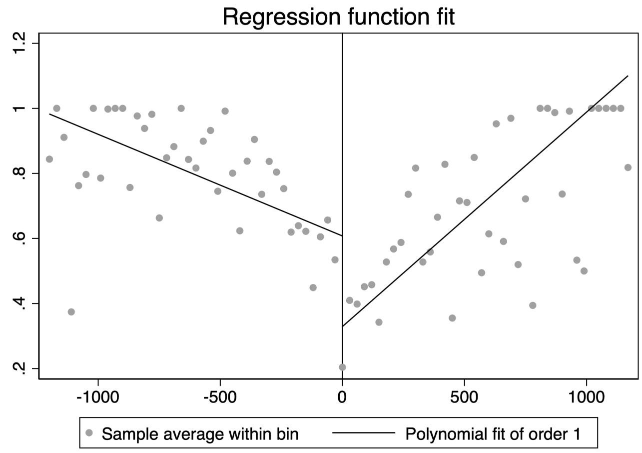
Table 7: Regression Discontinuity estimates on typing after achieving day 1 plans

	Time Window		
	30-seconds	60-seconds	90-seconds
Proportion typing	-0.274***	-0.255***	-0.075
	(0.101)	(0.087)	(0.108)
Obs. (left)	66,786	67,985	67,119
Obs. (right)	80,270	89,006	92,999
Clusters (left)	157	154	155
Clusters (right)	154	157	161
Bandwidth	339.9	368.3	422.3

Note: This table presents robust regression discontinuity estimates of the treatment effect on task choice at the moment participants achieve their day 1 planned typing duration. Standard errors clustered by participant in parentheses. The outcome variable is a binary indicator for choosing the typing task. Running variables are binned at 30-, 60-, and 90-second intervals around the achievement threshold. All estimates use triangular kernel with MSE-optimal bandwidth selection. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Visual inspection along with the results from 90-second window suggests this discontinuity it not all there is to know about participant behavior: While there is an initial drop in typing rate after achieving day 1 plans, participants eventually ramp up the time they spend at typing (see Figure 9). The positive slope on the right hand side illustrates this effect. Participants who achieve their day 1 plans earlier may in fact eventually get "bored" of the watching task as well - a behavior that shows how our efforts to create an addictive leisure task may have ultimately failed. These observations are coming from the about at least a third of all participants who were mostly watchers or alternaters in their day 1 types, but have increased their typing task consumption on day 2 (about respectively $(184 * 0.28 * 0.4)/184 = 11,2\%$ and $(184 * 0.54 * 0.4)/184 = 21,6\%$ of participants).

Figure 9: Linear fit for typing rate before and after achieving day 1 plan



Note: This figure plots the average typing rate across 30-second bins for participants as they approach their day 1 goal. The negative slope on the left hand side indicates participants working less and less as they approach the cutoff. The drop showcases the discontinuity where achiving the day 1 goal causes in typing behavior. The positive slope on the right hand side reveals another trend: As time goes on, participants tend to go back to the typing task and decrease their video consumption. This demonstrates participants' tendency to spend more time on the effortful typing task than originally planned.

4.6 Does Autoplay increase video consumption?

As participants were mostly focused on typing, they did paid less attention to the watching task, i.e, reducing its consumption value as evidenced by day 2 behavior. Consequently, in our setting experimenter demand effects is acting as a confound, in the sense of making more difficult to show statistically significant evidence in support of the hypotheses regarding the direction of deviations from the day 1 plans and the effects of the **Autoplay** condition. A lack of evidence in support of these objectives in our setting could be due not to a genuine failure of the corresponding hypotheses but rather to experimenter demand effects (Zizzo, 2010). With these in mind, we summarize variables of interest from the main 20-minute session across both conditions in Table 8. Two sample t-tests show no significant differences in any of the outcome variables. This suggests that **Autoplay** condition had no impact on the watching behavior observed in the experiment. Further, as established before, participants spend more time typing when compared to their Time Choice on day 1 (0.158, $t(183) = 8.92$, $p < 0.001$).

Table 8: Sample statistics for variables of interest

	Autoplay N = 79	**Control** N = 105	***p*** [A = C]
Time Choice			
Mean (SD)	0.48 (0.31)	0.44 (0.33)	0.410
Median [Min, Max]	0.50 [0.00, 1.00]	0.50 [0.00, 1.00]	
Proportion Typing			
Mean (SD)	0.65 (0.31)	0.58 (0.33)	0.173
Median [Min, Max]	0.72 [0.00, 1.00]	0.56 [0.00, 1.00]	
Nb. of sessions			
Mean (SD)	5.76 (4.98)	4.89 (5.09)	0.246
Median [Min, Max]	4.00 [1.00, 18.00]	3.00 [1.00, 28.00]	
Session length			
Mean (SD)	456.11 (383.22)	546.35 (411.53)	0.131
Median [Min, Max]	314.75 [68.17, 1212.00]	418.67 [44.93, 1314.00]	
Nb. of CAPTCHA submissions			
Mean (SD)	19.76 (12.11)	17.21 (11.62)	0.150
Median [Min, Max]	19.00 [0.00, 53.00]	15.00 [0.00, 45.00]	
Nb. of videos watched			
Mean (SD)	31.33 (26.34)	32.72 (26.39)	0.723
Median [Min, Max]	25.00 [0.00, 80.00]	34.00 [0.00, 77.00]	
Content rating			
Mean (SD)	6.96 (2.18)	6.62 (2.45)	0.327
Median [Min, Max]	7.00 [2.00, 10.00]	7.00 [0.00, 10.00]	

Note: This table presents key outcome variables from the Main Session. "Time Choice" refers to participants' stated preferences for typing on day 1 (proportion), while "Proportion Typing" represents the actual proportion of time spent on typing task during the 20-minute Main Session on day 2. Nb. of sessions is the number of typing or watching sessions participants completed and Session length represents the mean duration of individual sessions in seconds. Content rating reflects participants' evaluation of video content on a 0-10 scale. p-values are from two-sample t-tests comparing experimental conditions.

To formally test Hypothesis 1, i.e., effects of the **Autoplay** condition on time spent on typing, we run the regression specified in Equation 1. Our analysis reveals that the autoplay manipulation had no statistically significant effect on actual typing behavior. Table 9 presents results for typing proportion as the dependent variable. The coefficient on the autoplay treatment ranges from 0.12 to 0.20 across specifications but remains statistically insignificant in all models (p-values > 0.16). However, participants' day 1 plans emerge as a powerful predictor of day 2 behavior: a one standard deviation increase in planned typing time is associated with a 0.67 standard deviation increase in actual typing time ($p < 0.001$). Including day 1 plans increases the model's explanatory power dramatically, with R-squared rising from 1% to 46%.

Table 9: Effect of Autoplay Treatment on Typing Behavior

	Proportion Typing (z-score)		
	(1)	(2)	(3)
Autoplay	0.203	0.121	0.119
	(0.147)	(0.109)	(0.110)
Time Choice (z-score)		0.673***	0.651***
		(0.057)	(0.081)
Autoplay × Time Choice			0.057
			(0.110)
Constant	−0.087	−0.052	−0.053
	(0.100)	(0.076)	(0.076)
Obs.	184	184	184
R-squared	0.010	0.462	0.463
F-statistic	1.91	77.42	61.52

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Similarly, Table 10 examines the effect on video consumption, directly testing our primary hypothesis that autoplay would increase video watching. Across all specifications, the autoplay coefficient is statistically insignificant and even negative in the baseline model (-0.053, $p = 0.723$). This finding contradicts our theoretical prediction and suggests that the autoplay feature did not meaningfully alter participants' video consumption patterns. Again, day 1 plans prove to be the dominant predictor: participants who planned more typing watched significantly fewer videos ($\beta = -0.66$, $p < 0.001$), explaining 44% of the variance in video consumption.

Table 10: Effect of Autoplay Treatment on Video Watching Behavior

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	-0.053	0.029	0.031
	(0.149)	(0.112)	(0.112)
Time Choice (z-score)		-0.664***	-0.619***
		(0.056)	(0.079)
Autoplay × Time Choice			-0.114
			(0.105)
Constant	0.023	-0.012	-0.010
	(0.098)	(0.076)	(0.076)
Obs.	184	184	184
R-squared	0.001	0.440	0.443
F-statistic	0.13	73.06	61.53

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

For robustness, we provide the regression outputs from the unrestricted dataset in Appendix Table A.2 and A.3. Comparing the interaction models (column 3), we observe that the autoplay treatment effect on typing behavior decreases from 0.119 in the restricted sample to 0.055 in the unrestricted sample, while the autoplay treatment effect on video watching increases from 0.031 in the restricted sample to 0.132 in the unrestricted sample. However, the null results remain unchanged across both specifications. Because we cannot be sure these 40 participants actually engaged in watching videos, we base our conclusions on the restricted sample of 184.

To summarize, we observe two key patterns in the data that help explain our null findings. First, we observe contradictory results: the **Autoplay** condition increases time spent typing by 0.12 standard deviations (though statistically insignificant) while simultaneously increasing the number of videos watched by 0.03 standard deviations (also insignificant). If people type more, shouldn't they watch fewer videos? We identified a mechanical confound in our experimental design: the total time lost during video transitions. Participants in the **Control** condition, who had to manually click to start each new video, lost an average of 46.52 seconds due to these transition delays whereas those in the **Autoplay** condition lost only 1.08 seconds. This large difference of 45.45 seconds per session is statistically significant ($t(105) = 4.89$, $p < 0.001$). We find that this difference in time lost not watching videos plausibly explain the contradicting results and underscores the importance of looking at the unbiased outcome variable (i.e., number of videos watched) when evaluating the effects of the autoplay feature.

Second, assuming the actual effect size for the number of videos watched lie somewhere between the restricted and unrestricted samples, our power calculations based on pretests were insufficient: The required sample size to provide a two-sided test for Hypothesis 1b with the usual parameter values ($\alpha = .05$, $\beta = .2$) would be between 300 to 5000 per treatment, with common standard deviation as the root MSE from the standardized number of videos watched variable in the third regression. Other solutions to this power problem are design-related and involve extending the main-session duration to above 20 minutes, or nudging participants to mix more between tasks to avoid corner solutions (i.e., higher earnings when setting Time Choice between .25 and .75).

Table 11: Total Transition Time by Treatment Condition

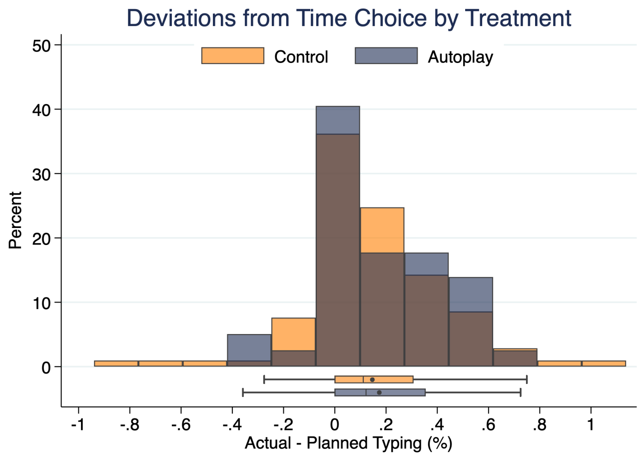
	Control	Autoplay	Difference
Total time lost	46.52	1.08	45.45***
	(9.27)	(0.65)	(9.29)
Observations	105	79	184
t-statistic			4.89

Note: Standard errors are in parentheses. Time is measured in total seconds lost to video transitions per session. The difference was tested using a two-sample t-test with unequal variances. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

4.7 Effects of autoplay on deviations from Time Choice

Does the **Autoplay** condition have an effect on deviations from day 1 Time Choice? To assess Hypothesis 2, we calculated the deviation by subtracting participants' planned typing proportion from their actual typing proportion (actual minus planned). Figure 10 shows the distribution of deviations from Time Choice across treatments, which appears right-skewed with positive values indicating participants typed more than originally planned. Both treatment groups systematically over-typed relative to their day 1 plans: Participants in the **Control** condition deviated by an average of 14.6 percentage points (SD = 27.6) while Autoplay participants deviated by 17.3 percentage points (SD = 23.6). However, a two-sample t-test reveals no statistically significant difference between means ($t = -0.694$, $p = 0.488$). The Kolmogorov-Smirnov test similarly finds no significant difference in distributions across conditions ($D = 0.085$, $p = 0.903$). These results indicate that while participants in both conditions systematically spent more time on the effortful typing task than they initially planned, the **Autoplay** condition did not significantly alter the magnitude or pattern of these deviations. Contrary to our hypothesis, we find no evidence that autoplay led participants to deviate more toward video watching behavior.

Figure 10: Distribution of Deviations from Time Choice by Treatment



Note: This figure shows the distribution of deviations from participants' day 1 time allocation plans, calculated as actual typing proportion minus planned typing proportion. Positive values indicate participants typed more than planned, while negative values indicate less typing than planned. The distributions are similar across treatment conditions, with both groups showing systematic over-typing relative to initial plans.

4.8 WTP for the commitment device (H3)

Hypothesis 3 explores whether participants would choose to turn Autoplay off when given the possibility, implying positive demand for a commitment device. We assess this demand by comparing the share of participants choosing to turn Autoplay on or off along with the associated bonus payments for each of the nine decisions in the price list. To quantify participants' willingness to pay (WTP) for Autoplay, we estimate the regression specification in Equation 2, where the dependent variable is the binary choice of selecting Autoplay (1) or turning it off (0), and the independent variable is the bonus amount in pounds from the MPL. The model is clustered at the participant level to account for 9 repeated decisions by the same individual.

Results are presented in Table 12. The coefficient on the bonus amount is 27.97 ($p < 0.001$), indicating that participants are highly sensitive to the financial cost of maintaining Autoplay. The WTP for Autoplay can be calculated as the negative ratio of the constant to the bonus coefficient: $-0.6263/27.9729 = -0.0224$ pounds, or approximately 2.24 pence. This suggests that participants value Autoplay at 2.24 pence on average and would be willing to forgo this amount to maintain access to the feature rather than have it turned off. In light of how participants operated in the decision-making environment, it appears that participants did not perceive a tradeoff from Autoplay in terms of increasing video consumption, but instead viewed it as facilitating leisure consumption before returning to the typing task.

Table 12: Willingness to Pay for Autoplay

	(1)
Bonus (£)	27.973***
	(6.277)
Constant	0.626**
	(0.199)
Obs.	468
Clusters	52
Pseudo R^2	0.658
Wald $\chi^2(1)$	19.86

Note: Logistic regression with standard errors clustered at the participant level in parentheses. Dependent variable is binary choice of selecting Autoplay (1) or turning it off (0). Bonus amounts range from -0.5 to 0.5 pounds. The pseudo R-squared (0.6584) indicates that the bonus amount explains a substantial portion of the variation in participants' choices. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

5 Discussion - integrate to introduction, redo and send

There are two ways to interpret our findings. We begin our discussion first with a simplistic understanding of the results and build up for a more nuanced explanation. The straightforward conclusion based on the lack of support for Hypothesis 1 is that Autoplay does not have a distinct impact on participants' actualized decisions when contrasted to active decisions. Combined with the lack of support for Hypothesis 3 on deviations from how long participants wanted to watch videos initially and the actual preference for Autoplay in Hypothesis 4 further supports the idea that Autoplay is a 'harmless' default when it comes to media control settings and does not provoke increased consumption. Our findings allow for this conclusion, and show instead that users actually prefer autoplay to active decisions when there is money at stake. This would contradict the presumed welfare benefits of the proposed ban in the SMART Act by the US Congress. However, this 'simple' interpretation of results does not consider the complexity of the issue at hand and leaves out a crucial design choice: The content of videos. To reassure, how the content was perceived by participants in our experiment was controlled by a survey question which asked them to rate the content of the videos. The median response was a 7 out of 10 rating across conditions for the content (see Table 8 for sample statistics).[12] Yet, the selection (manual) and the presentation (randomized order) of the content are clearly different from the algorithmically curated personalized experience participants are accustomed to have on daily basis. To have a clear understanding of the implications of the experiment, it is preferable to look at the larger context in which content and design together shape the digital environment where decisions are made.

In order to take into account this interplay between the content and the design, we rely on the insights from a recent model on media consumption (Kleinberg et al., 2022) where firms optimize over a large set of potential content and interface designs parametrized by their underlying features. Consumers optimize utility for their two selves: The impulsive and myopic one (self 1) and the forward-looking and thoughtful one (self 2). Content is consumed so long as self 1 derives some positive utility, and only after self 1 receives negative utility can system 2 make the decision to stop consuming

[12]Using Shapiro-Wilk tests we cannot reject the null that distributions are significantly different from a normal distribution for both autoplay ($W = 0.932$, $p < 0.001$) and control ($W = 0.933$, $p < 0.001$) treatments. A Mann-Whitney U test between treatments cannot reject the null that distributions are significantly different from one another ($U = 6894$, $p = 0.173$), although there is some indication (see Figure 11 for the density distribution) that participants in autoplay condition gave higher ratings to the content.

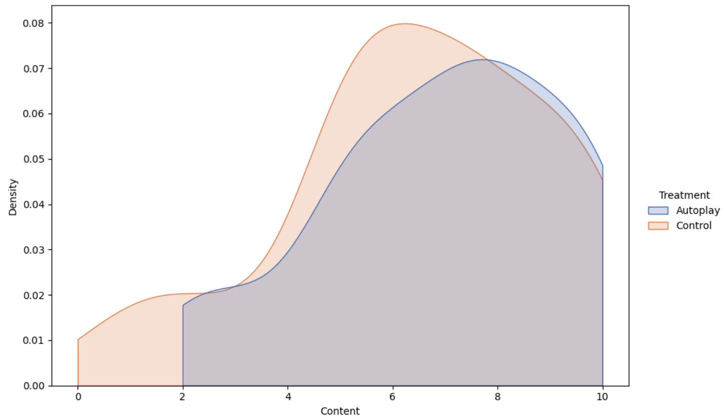


Figure 11: Density distribution of content ratings across treatments

content in case system 2 was already receiving negative utility. In this model, autoplay, a feature that facilitates seamless content consumption, has heterogeneous effects over different types of content. The intuition is as follows: If the existing content already is such that it has a low probability that utility for impulsive self 1 is larger than zero, autoplay has little impact on the user behavior. In other words, on a content landscape with low addictiveness, autoplay would have minimal impact on engagement. Applying this to our setting implies that the content provided for the experiment, although itself curated from social media, is much less interesting for the impulsive self 1 of participants.

Further, when a user is actively engaged and desires to consume more content, the autoplay feature aligns with the user's intent. This supports the preference for autoplay found in the MPL condition regarding Hypothesis 4, implying that participants were aware of their actual content consumption and desired to consume more content. This is entirely in accordance with the inference on the selection of videos not being tempting enough. In other words, because videos were not tempting, participants were in control of their consumption and could end sessions whenever their system 2 received negative utility from videos. In this setting, autoplay is actually helpful to participants in achieving their desired session length as it reduces decisions costs attached to it, fully aligned with previous the modelization of active decisions compared to defaults by Carroll et al. (2009). Putting the dual insights of the model together builds a coherent picture in terms of the policy implications of the present study, and we present convincing evidence that autoplay itself does not promote addiction in the absence of engaging content. Our findings go contrary to the proposed legislation which implicitly assumes all autoplayed content is equally addictive and welfare decreasing.

6 Conclusion

EXPERIMENTAL DEMAND EFFECTS AND SOCIAL IMAGE CONCERNS. THE EXPERIMENT IS SET AS A WORKING ENVIRONMENT. PARTICIPANTS PERCEIVED THE DECISION AS ONE TO WORK AS MUCH AS POSSIBLE.

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A Additional Figures and Tables

Additional Figures

Additional Tables

Table A.1: Comparison of main sample vs flagged participants

	Main Sample N = 184	**Flagged Sample** N = 40	***p*** [M = F]
Gender			
Female	89 (48.37%)	13 (32.50%)	0.068
Age			
Mean (SD)	42.29 (11.41)	42.85 (10.75)	0.768
Employment			
Employed (full time)	102 (55.43%)	27 (67.50%)	0.162
Employed (part time/self)	48 (26.09%)	8 (20.00%)	0.420
Not employed	34 (18.48%)	5 (12.50%)	0.366
Income			
Low (0-15)	56 (30.43%)	6 (15.00%)	0.048*
Middle (15-50)	103 (55.98%)	31 (77.50%)	0.012*
High (50+)	25 (13.59%)	3 (7.50%)	0.291
Marital Status			
Married/Partnership	95 (51.63%)	21 (52.50%)	0.921
Single	77 (41.85%)	17 (42.50%)	0.940
Previously married	12 (6.52%)	2 (5.00%)	0.719
Treatment Assignment			
Autoplay condition	79 (42.93%)	25 (62.50%)	0.025*

Note: This table compares demographic characteristics between the main sample (N=184) and participants flagged for potential inattention. Flagged participants reported engaging with videos but had mouse activity detected outside the experiment window, making it impossible to verify genuine engagement. p-values for binary outcomes are from two-sample tests of proportions; for continuous variables, from two-sample t-tests. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.2: Effect of Autoplay Treatment on Typing Behavior

	Proportion Typing (z-score)		
	(1)	(2)	(3)
Autoplay	0.136	0.055	0.055
	(0.133)	(0.108)	(0.108)
Time Choice (z-score)		0.595***	0.580***
		(0.061)	(0.089)
Autoplay × Time Choice			0.034
			(0.119)
Constant	−0.063	−0.026	−0.027
	(0.094)	(0.078)	(0.079)
Obs.	224	224	224
R-squared	0.005	0.357	0.357
F-statistic	1.04	50.28	35.00

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Table A.3: Effect of Autoplay Treatment on Video Watching Behavior

	Nb. of Videos Watched (z-score)		
	(1)	(2)	(3)
Autoplay	0.051	0.131	0.132
	(0.135)	(0.109)	(0.109)
Time Choice (z-score)		-0.590^{***}	-0.555^{***}
		(0.060)	(0.084)
Autoplay $\times$ Time Choice			-0.077
			(0.118)
Constant	-0.023	-0.061	-0.059
	(0.090)	(0.075)	(0.075)
Obs.	224	224	224
R-squared	0.001	0.347	0.348
F-statistic	0.14	48.91	34.21

Note: Robust standard errors in parentheses. All variables are standardized. Time Choice represents participants' planned typing proportion from day 1. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.